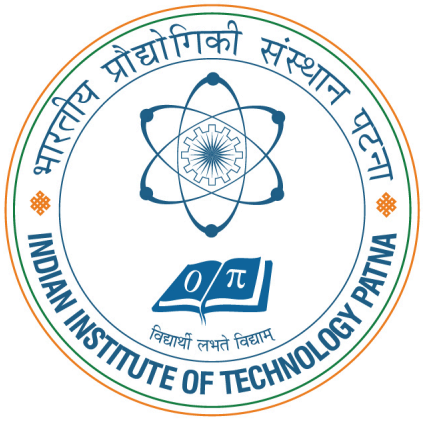
Computer Science & Engineering

Indian Institute of Technology Patna



PROJECT REPORT

C-Grade

Automated C-Code Grading System

Submitted by

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Submitted as part of the Mini Project / Lab Work

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C-Grade: An Automated, Rule-Driven Evaluator for C Programs

End-to-End Implementation Plan, Workflow Pipeline, Workload Distribution & Progress Report (agentic01)

- **Course:** Agentic AI Assignment (Phase 1 Deliverable)
- **Institution:** Indian Institute of Technology Patna
- **Date:** October 2026
- **Team Members:**
  - **Manas Gogna** — Roll No: 2611CS10
  - **Husain Kanchwala** — Roll No: 2611CS04
  - **Jay Agrawal** — Roll No: 2611AI09
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1. What We Are Building and Why (Project Overview)

Grading C programs by hand for a full class is slow, repetitive, and often uneven. Different evaluators check different things, and a single student program with an infinite loop or a bad pointer can freeze a basic grading script.

To solve this, we are building **C-Grade (AutoGrader)** — a complete, self-running grading system with a **Streamlit Web App (ui.py)** on the frontend and a **Python 3.11 Backend (agent.py)**, powered offline by a local AI model via **Ollama (qwen3.5-2b)**.

Our system is built on three clear promises:

1. **Same Code + Same Rules = Similar Grade Every Time:**
   Running the same C file with the same rules.json file always gives a similar score and feedback. We do this by using fixed test cases, counting CPU instructions (perf) instead of shaky wall-clock timers, and locking our local Ollama AI model to temperature = 0 and seed = 42.

2. **Rules Live Outside the Code (rules.json):**
   Teachers never need to edit Python code to change grading rules. Every metric, weight, threshold, timeout, and memory limit lives in rules.json and can be edited directly from the Streamlit Teacher Portal.
3. **Safe Sandbox Execution & Real Counter-Examples:**
   Every student binary runs inside a strict sandbox (1s timeout, 64 MB memory limit) so crashes and infinite loops are caught safely. When a check fails, the grader outputs a concrete counter-example showing the exact **INPUT**, **EXPECTED** output, what the program actually **GOT**, and **WHY** it failed.

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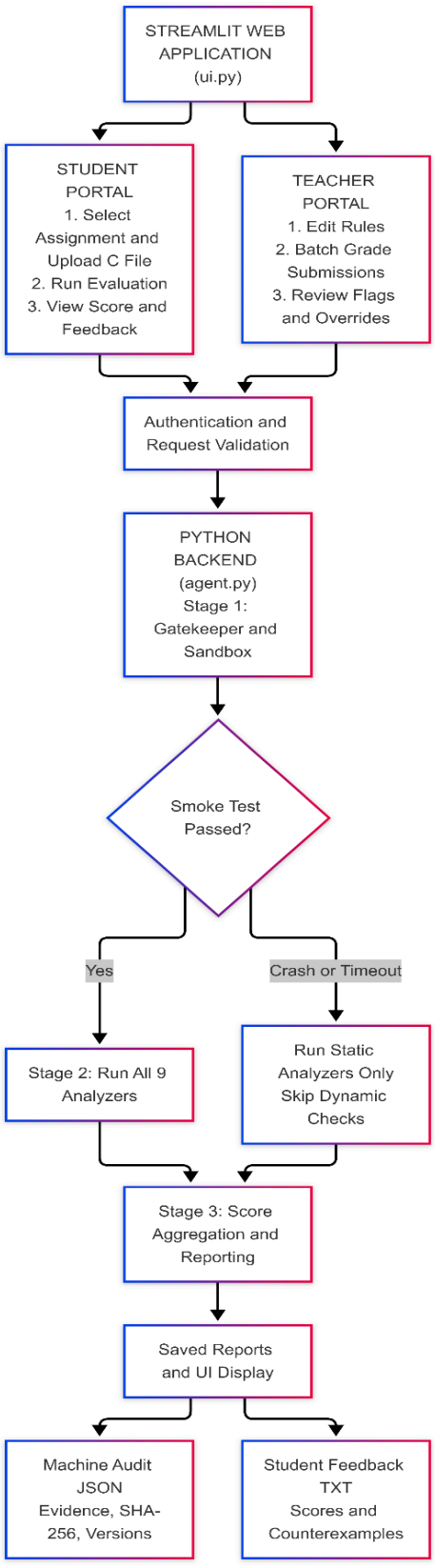
2. System Architecture (Streamlit Frontend + Python Backend)

2.1 How the Two Web Portals Work

- **Student Portal (Streamlit):**
  Students select an active assignment, upload their .c file (such as add.c), click **Evaluate My Code**, and immediately see their overall grade, a breakdown across all 9 metrics, compiler warnings, memory leak diagnostics, and counter-examples showing how to fix their bugs.
- **Teacher Portal (Streamlit):**
  Instructors visually configure rules.json (weights, limits, and thresholds), trigger **Batch Grading** across an entire folder of student submissions, check the **Plagiarism Triage Dashboard** for flagged pairs (similarity >= 0.85), inspect machine JSON logs, and override grades with audit logging.

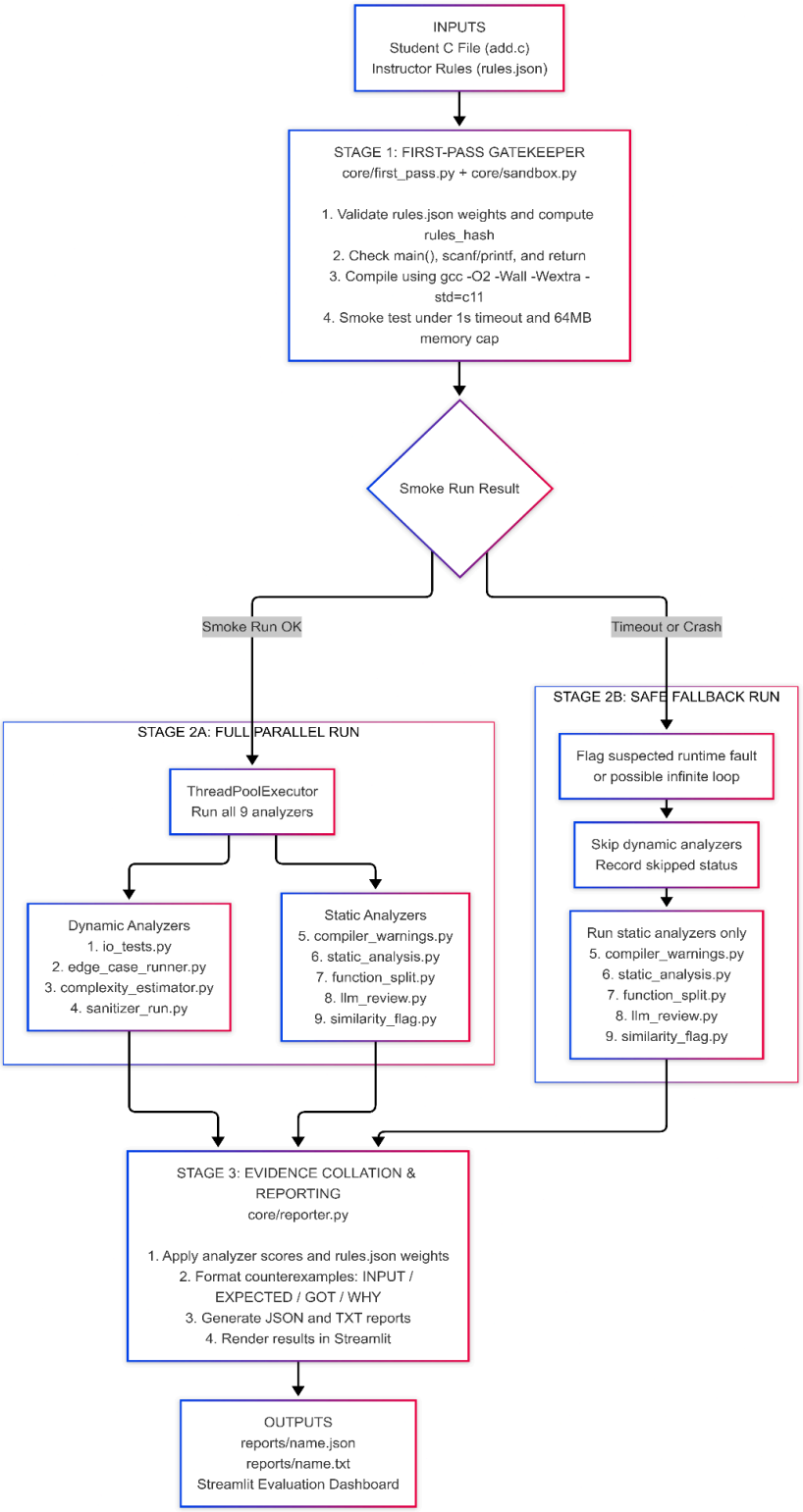
By unifying the student-facing submission workflow and the instructor-facing management interface within a cohesive Streamlit architecture, C-Grade minimizes friction across the entire evaluation lifecycle. Real-time feedback loops empower students to iteratively refine their code, while comprehensive batch controls and audit logs give instructors complete oversight and confidence in final grades.

2.2 Flowchart 1: Web Portal & Backend Architecture



3. End-to-End 3-Stage Grading Pipeline

3.1 Flowchart 2: How a Submission Travels Through the 3 Stages



4. The 9 Specialist Analyzers & Counter-Example Generator

Every analyzer uses a single shared Python function format: `evaluate(source_path, binary_path, params, limits) -> dict`

Metric	Module	How It Works (Tools & Method)	How We Keep It Reproducible
correctness	`io_tests.py`	Runs compiled binary on hidden .in/.out files in testset_dir.	Fixed inputs & 1s/64MB sandbox caps.
syntax	`compiler_warnings.py`	Parses gcc -fsyntax-only -Wall -Wextra -std=c11 warnings.	Pinned GCC version & fixed deduction.
structure	`static_analysis.py`	Uses cppcheck + lizard to check cyclomatic complexity & function length.	Deterministic AST rules.
modularity	`function_split.py`	Uses lizard AST check to count helper functions against min_functions.	Deterministic AST function count.
logical	`edge_case_runner.py`	Combines C boundary values (0, negatives, INT_MAX, INT_MIN) + Ollama (qwen3.5-2b) and checks against reference oracle.c.	Fixed templates + Ollama temperature=0, seed=42.
complexity	`complexity_estimator.py`	Runs scaled input sizes under perf stat instruction counts and fits complexity curve.	Hardware CPU instruction counts avoid OS timing noise.

Metric	Module	How It Works (Tools & Method)	How We Keep It Reproducible
memory_safety	`sanitizer_run.py`	Recompiles with -fsanitize=address,undefined (ASan/UBSan) and runs valgrind.	Deterministic sanitizer exit codes.
readability	`llm_review.py`	Local Ollama (qwen3.5-2b on port 11434) scores naming, comments, and formatting.	format="json", temperature=0, seed=42.
originality	`similarity_flag.py`	Normalizes C tokens and runs k-gram winnowing against corpus_dir submissions.	Exact Jaccard token similarity math.

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5. Real-Life Usefulness

1. **Automated Grading for Large University Courses (CS101 Labs):**
   Grades 400+ student C submissions in minutes with uniform feedback across all 9 metrics instead of slow, uneven manual TA grading.
2. **Technical Hiring & Coding Interview Screens:**
   Evaluates 500 job applicants not just on basic output, but on memory leaks (valgrind/ASan), time complexity (perf), and edge-case safety, giving recruiters a clear per-metric justification for every candidate.
3. **Plagiarism & AI-Copy Triage for Instructors:**
   Flags only submission pairs whose structural token similarity crosses 0.85, letting instructors review 10–15 suspicious pairs instead of manually checking 300 files.
4. **Self-Service Debugging Lab for Students:**
   Students can upload drafts to the Streamlit Student Portal before the deadline to catch memory bugs, compiler warnings, and corner-case failures on their own.

6. Workload Distribution (4 Team Members)

Member & Roll No.	Role	Assigned Modules	Core Responsibilities
Manas Gogna *(2611CS10)*	Coordinator & Sandbox Lead	`agent.py`, `core/first_pass.py`, `core/sandbox.py`, `test_tool.py`	Main orchestrator, JSON rule parser & SHA-256 hashing, Stage 1 Gatekeeper, 1s/64MB sandbox, and parallel ThreadPoolExecutor dispatch.
Husain Kanchwala *(2611CS04)*	Dynamic & Profiling Lead	`io_tests.py`, `edge_case_runner.py`, `sanitizer_run.py`, `complexity_estimator.py`	Hidden I/O runner, hybrid edge-case & counter-example generator with oracle.c, ASan/UBSan/Valgrind runner, and perf complexity estimator.
Jay Agrawal *(2611AI09)*	Static Analysis & Originality Lead	`compiler_warnings.py`, `static_analysis.py`, `function_split.py`, `similarity_flag.py`	GCC warning parser, cppcheck/lizard integration, AST helper modularity check, and k-gram token-winnowing plagiarism detector.
Adnan Hasan *(2611MC09)*	Local LLM, Reporting & UI Lead	`llm_review.py`, `core/reporter.py`, `ui.py (Streamlit)`, `rules/`	Local Ollama (qwen3.5-2b) setup, failure explanation synthesis, .json/.txt report generation, and Student/Teacher Streamlit portals.

7. Progress So Far & 3-Week Timeline

7.1 Progress So Far

- **Architecture & Unified Interface Locked:** Finalized the 3-stage pipeline and locked the evaluate(source_path, binary_path, params, limits) contract across all 9 analyzers.
- **Local Toolchain Verified:** Configured and tested Python 3.11, gcc, cppcheck, lizard, valgrind, perf, and local Ollama (qwen3.5-2b, temperature=0, seed=42).
- **Baseline Benchmark Prepared:** Set up add.c, reference oracle.c, add_rules.json, and the standalone analyzer test harness (test_tool.py).

7.2 3-Week Timeline

Week	Milestone Focus	Key Deliverables	Lab Checkpoint
Week 1 *(Oct 16 – Oct 22)*	Core Skeleton & Static/Dynamic Base	Submit agentic01 report; initialize GitHub repo (c-grade); build first_pass.py, sandbox.py, test_tool.py, io_tests, and sanitizer_run.	Demo Stage 1 Gatekeeper, sandbox, and initial static/dynamic test runs on add.c.
Week 2 *(Oct 23 – Oct 29)*	Full Analyzer Suite & Parallel Engine	Complete compiler_warnings, static_analysis, function_split, edge_case_runner, complexity_estimator, similarity_flag, and llm_review; integrate parallel dispatch.	Demo all 9 analyzers running in parallel with automatic counter-testcase generation.
Week 3 *(Oct 30 – Nov 5)*	Streamlit UI, Reporting & Final Video	Build Streamlit ui.py and reporter.py; verify 100% score reproducibility; record 15–20 min YouTube demo and prepare final presentation.	Submit GitHub repo, final report, and YouTube demo video.